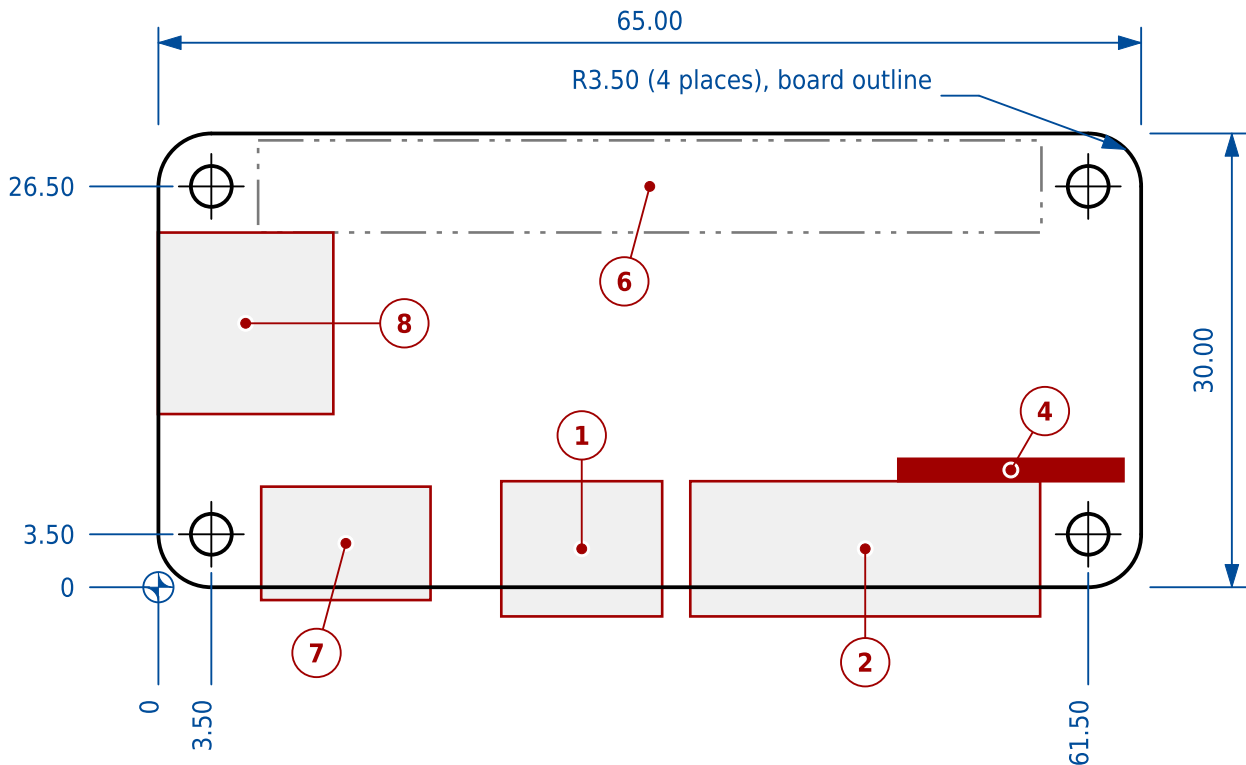




## HOLE SCHEDULE

| ID | X mm  | Y mm  | DIA mm | KEEPOUT mm |
|----|-------|-------|--------|------------|
| H3 | 3.50  | 3.50  | 2.70   | not given  |
| H4 | 61.50 | 3.50  | 2.70   | not given  |
| H1 | 3.50  | 26.50 | 2.70   | not given  |
| H2 | 61.50 | 26.50 | 2.70   | not given  |

## FEATURE SCHEDULE

| #  | FEATURE                                           | X EXTENT mm    | Y EXTENT mm    |
|----|---------------------------------------------------|----------------|----------------|
| 1  | USB-C J3, JTAG and console                        | 22.68 to 33.32 | -1.93 to 7.01  |
| 2  | USB-C J5 and J4, to the FPGA, 2 on 12.50 mm pitch | 35.18 to 58.32 | -1.93 to 7.01  |
| 3  | Ethernet RJ45 - not on this board                 | -              | -              |
| 4  | User LEDs LED0-LED4, 5 on 2.98 mm pitch           | 48.94 to 63.83 | 7.01 to 8.49   |
| 5  | User LEDs, tri-colour - not on this board         | -              | -              |
| 6  | GPIO header J1, 2x20, not fitted                  | 6.59 to 58.41  | 23.45 to 29.55 |
| 7  | GPD1 video connector J2                           | 6.80 to 18.00  | -0.85 to 6.65  |
| 8  | microSD card socket J6                            | -0.03 to 11.57 | 11.45 to 23.45 |
| 9  | Third USB port - not on this board                | -              | -              |
| 10 | Fourth USB port - not on this board               | -              | -              |
| 11 | Status LEDs, debug controller - not on this board | -              | -              |

## LEGEND

|  |                                      |
|--|--------------------------------------|
|  | Board outline                        |
|  | Component body, scheduled            |
|  | Position not fitted on this revision |
|  | Dimension, extension and leader      |

## NOTES

- Viewed from the component side.
- Hole IDs are the KiCad reference designators.
- The mounting holes are the Raspberry Pi Zero pattern: 58.00 x 23.00 mm, 3.50 mm in from each edge of a 65 x 30 board. They are drilled 2.70, the bottom of the Pi Zero drawing's 2.75 +/-0.05 band; the corners are R3.50 where the Pi Zero's are R3.00.
- No Pmod host and no Ethernet jack.
- The unfitted GPIO position is a Raspberry Pi 40-pin header: 1.00 mm holes on a 2.54 mm grid, with the Pi's 5 V, 3V3 and first GPIO pins in the Pi's places and pin 1 at (8.38, 25.23), in the row farther from the board edge.
- The two user buttons are on the underside, centred at (46.40, 15.45) and (46.40, 18.85); a plate under the board has to clear them.
- The board file has been re-annotated since the mass-production release. Nothing drawn here moved, but at d67bb758 the programming port is J5 and the user LEDs run D1 to D5 left to right; the designators here are the ones the sold boards were fabbed with.
- Assembled envelope, connector overhang included: 65.03 x 31.93 mm.

## SOURCES

- KiCad board file: [https://github.com/cheyao/icepi-zero/hardware/v1.3/icepi-zero.kicad\\_pcb@6e4aaba2](https://github.com/cheyao/icepi-zero/hardware/v1.3/icepi-zero.kicad_pcb@6e4aaba2) (tag v1.3) - title block: Icepi Zero rev v1.3, dated 2025-07-22; Chengyin Yao (cheyao). Solderpad Hardware Licence 2.1.
- Board revision: <https://github.com/cheyao/icepi-zero/releases/tag/v1.3> - the v1.3 tag, "Final mass production files": the revision made in quantity and sold. The same board file later in its history, at d67bb758, is read too and agrees on every position drawn here.
- Board size: <https://github.com/cheyao/icepi-zero/blob/v1.3/README.md> - quoted at this tag: "Icepi Zero is an FPGA development board in the popular Raspberry Pi Zero form factor"; the board file gives 65.00 x 30.00 mm.
- Assembly: <https://github.com/cheyao/icepi-zero/blob/v1.3/hardware/v1.3/production/bom.csv> - the production bill of materials at the same tag lists no 2x20 header: the GPIO position is not fitted.
- Form factor compared: <https://datasheets.raspberrypi.com/rpizero/raspberry-pi-zero-mechanical-drawing.pdf> - Raspberry Pi Zero, RPI-ZERO-V1\_2 of 2015-09-23: 65 x 30 mm, corner radius 3.0, 4x M2.5 mounting holes drilled to 2.75 +/-0.05, 58 x 23 apart and 3.5 in from each edge.

## ICEPI ZERO

v1.3, Raspberry Pi Zero form factor

|                  |                                                |                 |       |
|------------------|------------------------------------------------|-----------------|-------|
| DRAWN            | VERSION                                        | UNITS           | SCALE |
| generated        | v0.0-400-g80d77dd                              | mm              | 2:1   |
| SIZE             | SHEET                                          | DRAWING NO      | REV   |
| A3               | 1 OF 1                                         | FPGA-ICEPI-ZERO | A     |
| MATERIAL         | GENERAL TOLERANCE                              |                 |       |
| PCB, 1.6 nominal | edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08 |                 |       |